

Exothermic reaction

- transfers thermal energy to the surroundings leading to an increase in temperature of the surroundings.

Endothermic reaction

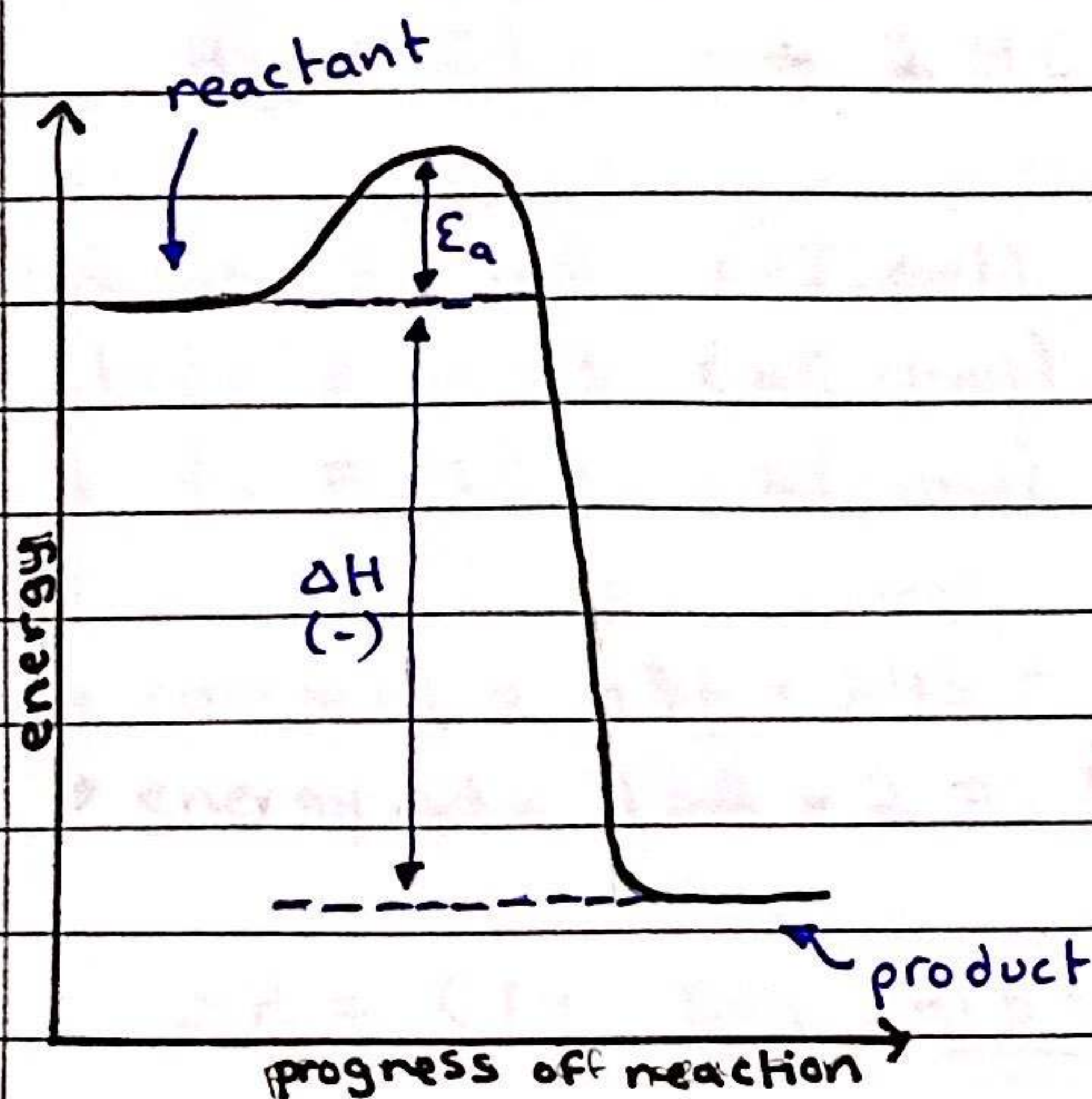
- takes in thermal energy from the surroundings leading to a decrease in the temperature of the surroundings.

Enthalpy change

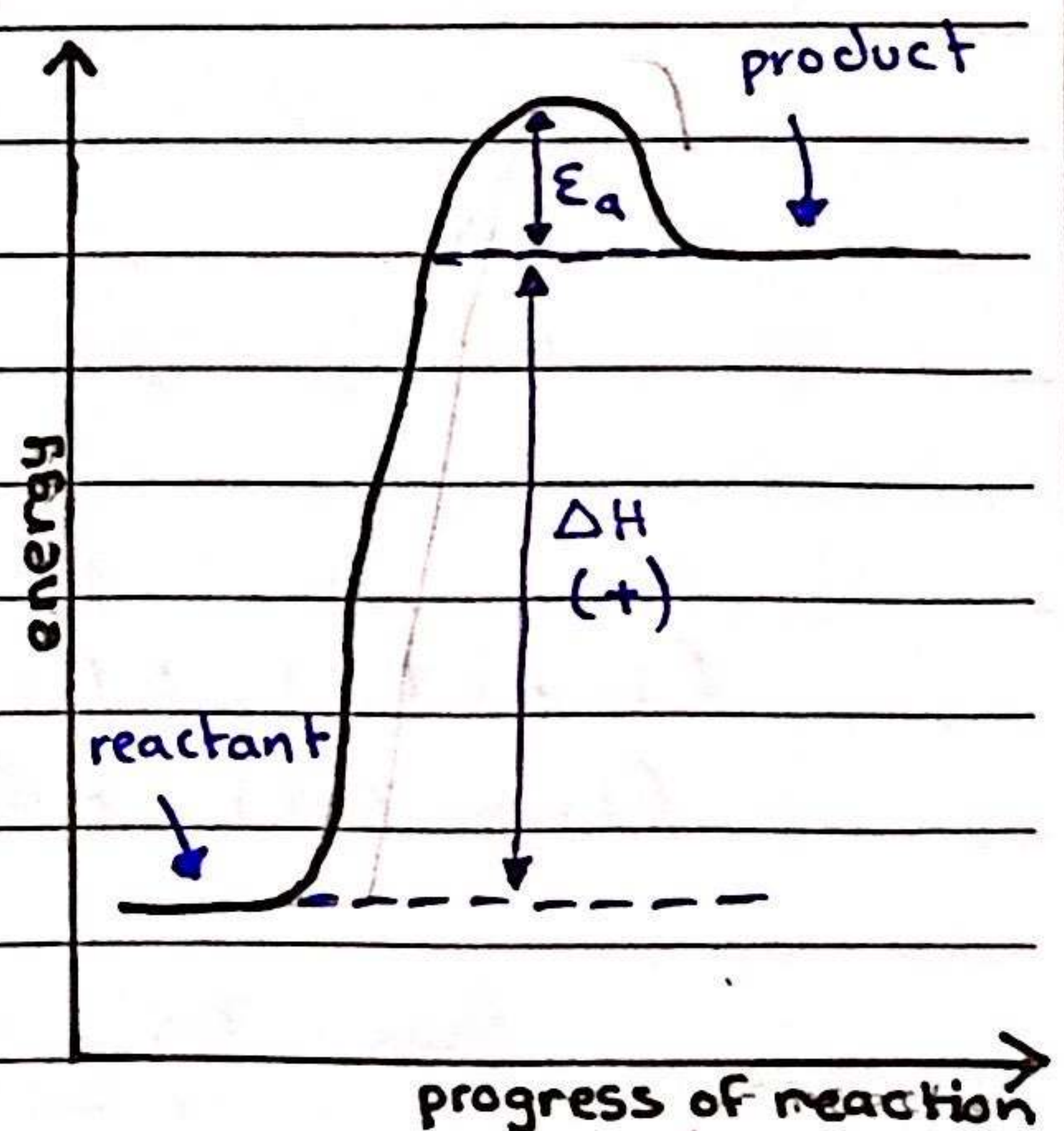
- transfer of thermal energy (ΔH)
- is negative for exothermic reactions
- is positive for endothermic reactions

Activation energy

- minimum energy that colliding particles must have to react (E_a)



[Exothermic]



[Endothermic]

* bond breaking is an endothermic process

* bond making is an exothermic process

→ if more energy is absorbed to break bonds than is released to form new bonds, the reaction is endothermic

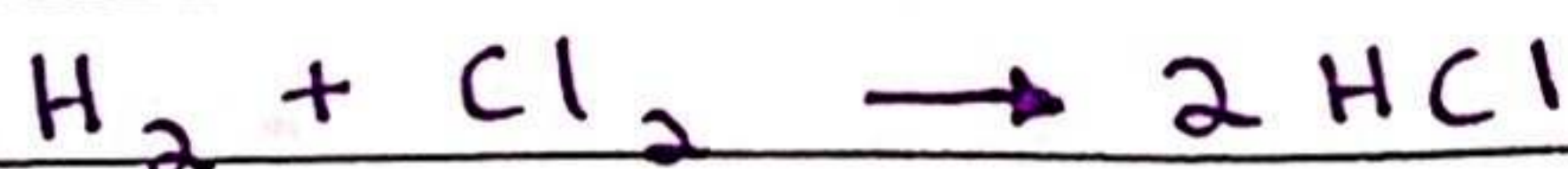
→ $[\Delta H]$ is positive since the products have more energy than the reactants

→ if more energy is released to form new bonds than is absorbed to break bonds, the reaction is exothermic

→ $[\Delta H]$ is negative since the products have less energy than the reactants

Calculating enthalpy change (ΔH) :

• $\Delta H = \text{energy taken in} - \text{energy given out}$



$$\text{H}-\text{H} = 436 \text{ (kJ/mol)}$$

$$\text{Cl}-\text{Cl} = 243 \text{ (kJ/mol)}$$

$$\text{H}-\text{Cl} = 432 \text{ (kJ/mol)}$$

$$\rightarrow \text{energy in} = 436 + 243 = 679 \text{ (kJ/mol)}$$

$$\rightarrow \text{energy out} = 432 \times 2 = 864 \text{ (kJ/mol)}$$

$$\Delta H = 679 - 864 = \underline{\underline{-185 \text{ (kJ/mol)}}}$$

* enthalpy change is negative hence it is an exothermic reaction